

NGUYEN HOANG ANH QUAN (FINN)

BACKEND DEVELOPER - GOLANG & NODE.JS

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PROFESSIONAL SUMMARY

Final-year IT student specializing in high-performance Backend Development with Golang and Node.js. Demonstrated expertise in Clean Architecture, multi-factor security mechanisms (JWT single-use rotation, RFC 6238 TOTP 2FA, FIDO2/WebAuthn Passkeys), database transaction reliability (MySQL, PostgreSQL, Redis 7), and automated testing with over 301 unit & fuzz test cases. Passionate about solving real concurrency & distributed systems challenges.

CORE TECHNICAL SKILLS

Backend & Architecture: Golang (1.26), Gin, GORM, Node.js (22 LTS), TypeScript, Clean Architecture, RESTful API, DTOs
Databases & Caching: MySQL 8 (Indexing & Transactions), PostgreSQL 16, Redis 7 (Sessions, Sliding Limits, Token Blacklist)
Security & Auth: JWT Rotation & Family Revocation, TOTP 2FA (RFC 6238), WebAuthn Passkeys, BCrypt (72B cap), Argon2id
DevOps & Testing: Docker & Docker Compose, GitHub Actions (7 automated CI jobs), Unit & Fuzz Testing (301 tests), slog, Prometheus

FEATURED ENGINEERING PROJECTS

- FinnApiGo - Authentication & Multi-Factor Security Engine in Go***Live on Render | Golang, Redis 7, MySQL 8*
 - Architected a production-ready authentication engine in Go strictly following 5-layer Clean Architecture with zero Gin leak.
 - Implemented JWT Access Token (15m) paired with single-use Refresh Token rotation stored as SHA-256 hashes in Redis 7.
 - Engineered Token Family Revocation: automated intrusion detection revoking entire session family upon replay attack detection.
 - Built sliding-window rate limiting with IPv6 /64 subnet collapse mitigating botnet brute-force attacks with HTTP 429.
 - Created dynamic TOTP 2FA provisioning (RFC 6238) with AES-256-GCM encrypted recovery codes and FIDO2/WebAuthn Passkeys.
 - Automated comprehensive 7-job CI pipeline: 301 unit tests, 3 fuzzing targets, gosec security scans, and Trivy CVE scanning.
- VovinamClubManagement - Martial Arts Management System***Thesis Capstone | Node.js, TS, PostgreSQL 16*
 - Engineered backend REST API handling multi-tenant martial arts club operations: member grading, attendance, and finances.
 - Designed strict Role-Based Access Control (RBAC) matrix for Admin, Coach, Student, and Cashier with object-level validation.
 - Integrated dynamic VietQR payment automation with webhook HMAC-SHA256 signature verification & idempotent ledgers.
 - Modeled normalized relational schemas on PostgreSQL 16 with UUID primary keys and automated Docker schema migrations.
- JakeAI - Dedicated Interactive AI Assistant Microservice***Independent Go Backend | Gin, Reverse Proxy*
 - Developed standalone Go microservice acting as conversational AI assistant to guide recruiters through portfolio systems.
 - Implemented lightweight in-memory knowledge store with fallback heuristics ensuring 99.9% uptime even during cold boots.

TECHNICAL WRITE-UPS & DEBUGGING CASE STUDIES

- Zero-Downtime DB Migration:** Resolved production schema deadlocks via 3-phase Expand/Contract pattern & Goose transaction guards
- Sudo Mode Re-Authentication:** Designed 5-minute cryptographic step-up tokens in Redis to secure sensitive account operations.
- WebAuthn RPID Domain Binding:** Fixed FIDO2 signature validation across localhost/staging/prod via dynamic origin whitelist middleware.
- Token Rotation Race Defense:** Eliminated multi-tab false-positive replay alerts via Redis Lua atomic single-use exchange.

EDUCATION & LANGUAGES

Bachelor of Information Technology (Final-Year Student)*Expected Graduation: 2026*
Focus: Software Engineering, Distributed Systems, Database Management, Network Security
Languages: Vietnamese (Native), English (Technical reading, documentation & API specifications)